# Research on Pharmacist-led Science Communication for Drug Package Insert Reform

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**Abstract:** To improve patients' healthcare experience and explore drug package insert reform, the research team investigated the problems of traditional drug package inserts. Based on the findings, the team integrated pharmacy science popularization resources to build a new visualized drug package insert and service model: science popularization videos are promoted through a fission communication model, and a self-developed "One Drug, One QR Code" system presents the drug science popularization videos as QR codes on prescription slips. By simply scanning the code, patients can quickly access concise, easy-to-understand visualized drug package inserts with real-person demonstrations and obtain individualized pharmaceutical services. This model has converted more than 150 drug package inserts, benefiting more than 850,000 people; it has been widely reported by many authoritative media outlets represented by Xuexi Qiangguo and has won national-level and municipal-level industry case awards. The research shows that this low-cost, easily replicable approach positively promotes the exploration of drug package insert reform, and that visualized package inserts and innovative pharmaceutical service models enable patients to "take their medicines with full understanding and use them with peace of mind".

**Keywords:** Drug package insert; Visualization; Internet Plus; Pharmaceutical services; Pharmacist-led science popularization

The report to the 20th National Congress of the Communist Party of China called for "giving strategic priority to ensuring the people's health and improving policies on promoting public health", and made comprehensive arrangements for "advancing the Healthy China Initiative". To popularize healthy lifestyles and raise the overall health of the population, the State Council has successively issued programmatic documents such as the *Notice of the State Council on Issuing the "Healthy China 2030" Planning Outline* and the *Healthy China Action (2019–2030)*. These documents put forward the strategy of "promoting Internet Plus precision health science popularization" and set out a series of requirements for further preventing and controlling major diseases and improving public health.

In May 2023, the National Health Commission and the National Administration of Traditional Chinese Medicine issued the *Notice on Launching the Thematic Campaign to Improve the Medical Care Experience and Enhance Patient Experience*, which set explicit requirements for further optimizing service processes, improving the medical care experience, improving work style and efficiency, and promoting the high-quality development of hospitals.

In October 2023, the National Medical Products Administration (NMPA) issued the *Work Plan for the Pilot Reform of Age-Friendly and Accessible Drug Package Inserts* (hereinafter "the *Plan*"), which explicitly requires optimizing the management of drug package inserts and encourages exploring ways to solve the problems of package inserts being hard to understand and hard to read. As one of the pilot provinces, Jiangsu issued a notice on detailed local implementation based on the *Plan* and included this work in the *Key Points of Drug Regulatory Administrative Approval Work in Jiangsu Province for 2024*.

The drug package insert is a statutory document that sets out important information about a drug and is the legal guide for selecting and using medicines. Every sentence in a package insert is supported by scientific data and cannot be modified lightly. To comply with regulatory requirements and to provide healthcare professionals with detailed drug information, package inserts must contain as much content as possible, so within the limited space of an insert, manufacturers can only reduce the font size. This has given rise to the industry-wide phenomenon of package inserts with "characters as small as ants" and "a single grain of rice can cover four characters". As a result, patients find it difficult to understand the usage and precautions described in package inserts, which is a common pain point in patients' experience of seeking medical care.

For most patients, the technical terminology and complex information in package inserts also form a barrier that makes it difficult for them to quickly and accurately understand important information such as indications, dosage and administration, adverse reactions, and contraindications. In elderly patients, memory and logical thinking gradually decline with age, which further aggravates their reading difficulties in understanding package inserts.

At present, the graphic design of package inserts is not intuitive enough, and patients often cannot accurately grasp how to use a drug and the related precautions. This can easily lead to misuse, such as taking a drug intended for external use by mouth, causing unnecessary health risks. In addition, most package inserts lack detailed descriptions of practical usage, such as whether rinsing is needed again after a sitz bath with a lotion, the amount required for each use (measured in bottle caps), and the volume of each bottle cap. This missing information causes patients many difficulties in use.

It can thus be seen that traditional drug package inserts no longer meet the public's current healthcare needs. We urgently need a new type of package insert that enables patients to understand the key information on drug use conveniently and intuitively.

At present, the package insert reform under the NMPA *Plan* focuses on encouraging marketing authorization holders to take part in the pilot and to provide large-print, electronic, or simplified versions of package inserts. However, merely enlarging the font size, although it solves the problem of "being able to see clearly" to some extent, does not address the core issue—namely, how to enable patients to "understand" the package insert and thereby ensure that they "use the drug well"[1].

Pharmacists are healthcare professionals who provide pharmaceutical technical services and are also the gatekeepers of patients' rational drug use. In their daily work they repeatedly explain drug usage to patients and have accumulated extensive experience; for administration methods that patients find difficult to understand, they have developed mature ways of explaining that allow patients to grasp them simply and clearly. Therefore, in the course of package insert reform, the professional strengths and accumulated experience of pharmacists can make them active participants in and promoters of the reform.

Pharmacy science popularization is an activity in which pharmacists disseminate scientific knowledge in the field of pharmacy to the public by means of health science popularization, with the aim of improving the public's medication-related health literacy. Through pharmacists' interpretation of package inserts, inserts that are closer to patients' needs and easier to understand can be developed. This innovative measure will significantly improve medication adherence and safety, improve patients' healthcare experience, enable more residents of Wuxi to "take their medicines with full understanding and use them with peace of mind", and provide valuable practical experience and directions for exploration for drug package insert reform in China[2].

# I. Research Methods

## (I) Production Process of Visualized Drug Package Inserts

1. Establishing the research team. Since its founding in 2020, the Anxin Pharmacist team has brought together a group of professionals in pharmacy, more than 70% of whom hold master's or doctoral degrees. Supported by the platforms of a Wuxi Key Medical Specialty, a National Health Commission Clinical Pharmacist Training Base, and a Pharmacy Science Popularization Base of the Jiangsu Pharmaceutical Association, the team includes 14 specialist clinical pharmacists and traditional Chinese medicine pharmacists.

2. Establishing a review mechanism for science popularization scripts. Using their professional pharmaceutical knowledge, the pharmacists distill the content of package inserts and convert it into popular science scripts that are easy to understand. The review process is rigorous: an initial review is carried out by a pharmacist with an intermediate professional title or above, followed by a final review by a pharmacist with the title of associate chief pharmacist or above, to ensure that the content is accurate and appropriate. The edited and polished science popularization videos are then submitted to a chief pharmacist for review to ensure both their scientific rigor and their appeal to viewers; once approved, they are published and promoted through internet platforms.

3. Producing science popularization videos. Scripts that have passed the rigorous review are handed over to dedicated multimedia pharmacists for production. According to the theme of each video, they style the presenting pharmacists in terms of costume, props, and make-up, and use shooting techniques such as fixed, push-in, pull-out, pan, and tracking shots to ensure that the camera language communicates effectively. Once the footage has been judged satisfactory on site, the multimedia pharmacists carry out post-production with professional video-editing software such as Premiere and Final Cut. During editing, they also edit the audio, adjusting the timbre of the

voices and adding background music to enhance the realism and atmosphere of the footage. Special effects, such as entry and exit effects, transition effects, and subtitle effects, further enrich the visual effect and better match the visual rhythm to the sound, giving patients a more intuitive and vivid drug science popularization experience[3].

### (II) Internet-Enabled Promotion of Visualized Package Inserts

1. Promotion through a fission communication model. The carefully edited and designed package insert videos are released through the Anxin Pharmacist science popularization platform, which is dedicated to popularizing drug safety knowledge. The platform adopts a fission communication model, breaking through the one-way, linear dissemination of traditional science popularization platforms. With fission communication, every user who watches a video can become a disseminator and share it with more people, so that the content spreads rapidly.

2. Cross-platform distribution of visualized package inserts. To further expand the influence of the videos, we will also promote them on the Cloud Science Popularization Day platform hosted by the China Association for Science and Technology and on Xiaohongshu. Both are influential internet science popularization platforms with very large user bases. We will push the videos to these platforms simultaneously to achieve a second round of fission of the content. In this way, the visualized package insert videos can be seen by more people, and drug safety knowledge will be disseminated more widely.

### (III) Visualized Package Inserts Empowering Prescription Slips

1. Developing the "One Drug, One QR Code" system. The research team planned to develop a "One Drug, One QR Code" system that matches each drug's generic name and the web link of its visualized package insert with the hospital's existing drug catalogue by generic name. When patients collect their medicines at the hospital, the QR code corresponding to each drug will be clearly printed on their prescription slips; with one quick scan on a mobile phone, patients can watch an easy-to-understand visualized package insert that can be viewed repeatedly. This innovative measure not only delivers science popularization content precisely to its target users but also offers an interactivity that traditional science popularization lacks.

2. Ensuring pharmaceutical services throughout the medication process. For patients who encounter problems while watching the visualized package inserts or who have further health needs, we will provide a QR code for the internet pharmacy clinic on the prescription slip. By simply scanning the code, patients can receive professional, detailed, and individualized pharmaceutical services at the pharmacy clinic of the internet hospital. For discomfort occurring after taking a medicine, we have also provided an event-reporting QR code on the prescription slip; after scanning it, patients can report the situation to pharmacists by completing a questionnaire. Pharmacists follow up each piece of feedback by telephone, determine whether an adverse drug reaction (ADR) has occurred, give individualized management advice, and report the ADR.

## II. Research Outcomes

To date, more than 150 visualized package inserts have been produced, mainly covering 5 categories: pediatric medicines, menopause medicines, gynecological external-use lotions and suppositories, antineoplastic drugs, and Chinese patent medicines. Since going online in July 2023, they have received more than 850,000 views in total, with a patient satisfaction rate of 100%. The visualized package inserts have been widely recognized in Jiangsu Province and were reported as a 2023 "World Pharmacists Day" demonstration project of the Jiangsu Provincial Health Commission. In addition, many authoritative media outlets, including Xuexi Qiangguo, *Health News*, Healthy Wuxi, Wuxi News Radio, Xibao Media, *Wuxi Daily*, *Yangtse Evening Post*, and Purple Cow News, have run special reports on this work. The project was also rated an Outstanding Project of the Wuxi Municipal Health Commission's 2023 "Hundred Health Micro-Initiatives", and was selected as the Best Case of the Annual Brand Activity in the 2023 *Health News* "Season 6 Search for Innovative Publicity Cases in the Health Industry".

## III. Research Significance

The exploratory work of this case on drug package insert reform is low-cost and easy to replicate and promote, and it has positive effects in three dimensions: for society, for patients, and for pharmacists.

### (I) Low Cost and Easy to Replicate and Promote

Drawing on their professional strengths, pharmacists summarize the key content of traditional package inserts and supplement the key points of use, breaking with the rigid layout and obscure content of traditional inserts. In today's boom of short videos, a pharmacist needs only a mobile phone and short-video editing software such as CapCut to create easy-to-understand videos explaining drugs. With the powerful reach of the internet, the videos can spread

quickly and widely across the major short-video platforms, greatly expanding the population that benefits. Because the server resources of the short-video platforms are used, all that is needed is to convert the web links into QR codes and match them with the drug names, which saves the considerable cost of deploying servers and maintaining websites. Patients do not need to download a dedicated app either: with a common app such as WeChat, they can simply scan the QR code to watch the visualized package insert and obtain individualized pharmaceutical services. This realizes "scan with your phone, take a pharmacist home" and creates a win-win situation in which the hospital reduces costs and improves efficiency while patients enjoy convenience and peace of mind.

(II) Improving the Health Level of Society

Optimizing and improving the management of drug package inserts, meeting the needs of different patients, and improving the quality, substance, and accessibility of pharmaceutical services are key links in promoting rational drug use and ensuring medication safety. By improving the readability and comprehensibility of package inserts, visualized inserts reduce the health risks caused by elderly patients misunderstanding how to use their medicines, as well as the unnecessary medical interventions and medical expenditure caused by medication errors, thereby optimizing the allocation of medical resources. Diversified channels of dissemination increase the public's exposure to and understanding of health information, which has a positive effect on raising the overall health level of society.

(III) Increasing Patients' Medication Safety

Small as it is, the drug package insert is of undeniable importance. Accurate and safe medication use bears directly on the health of countless families; enabling everyone to understand the insert "clearly and thoroughly" gives patients reassurance and peace of mind. Clear, easy-to-understand package inserts can improve medication safety and help ensure therapeutic effect, and they can also effectively reduce patients' psychological and cognitive burden during medication. Through the internet pharmacy clinic, patients can receive individualized pharmaceutical services at home, which improves their overall healthcare experience. Meanwhile, the feedback mechanism for medication experiences enables patients to take a more active part in managing their own health and strengthens their awareness of self-care.

(IV) Enhancing the Image of Pharmacists

By taking part in producing and interpreting visualized drug package inserts, pharmacists not only improve their own professional pharmaceutical skills but also step out from behind the scenes to center stage, becoming an indispensable part of public health education and drug safety publicity, so that more patients come to know, recognize, and trust pharmacists. The medication education and timely pharmaceutical service interventions that pharmacists provide in this process safeguard patients' medication safety and health, while also strengthening pharmacists' professional pride and sense of self-worth and fulfilling the purpose of serving the people.

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